with the cleaning technologies available to ensure both minimum biofouling growth as well as reducing the risk of damage to the AFC and the potential release of harmful waste substances to the environment.

.6 Maintenance: The lifespan of an AFS should be considered in combination with dry-docking schedules. AFC lifespan and lifetime of MGPS (e.g. anodes) should exceed the period between dry-dockings.

.7 Legal requirements: In addition to the AFS Convention, any national or

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SOURCE: MEPC.378(80).pdf
with dry-docking schedules. AFC lifespan and lifetime of MGPS (e.g. anodes) should exceed the period between dry-dockings.

.7 Legal requirements: In addition to the AFS Convention, any national or regional regulatory requirements, if relevant, should be considered in the selection of AFS. This may apply to release of chemicals from MGPS and the AFS.

Installing the AFS

6.4 Installing an AFS in hull and niche areas should be in accordance with the manufacturer's guidance.

6.5 Niche areas are particularly susceptible to biofouling growth. Care should be taken in surface preparation and application of any AFC to ensure adequate adhesion and coating thickness. Particular attention should be paid to corners, edges, pipes, holding b rackets and bars of gratings. Corners, edges and welded joints should be smooth and coated with adequate coating thickness to optimize system effectiveness. Additionally, for such areas, it is recommended to apply a touch up to ensure film thickness or a higher-grade AFC.

6.6 A non-exhaustive list of recommended measures for installation of an AFS in niche areas is as follows:

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SOURCE: MEPC.378(80).pdf
recommended to apply a touch up to ensure film thickness or a higher-grade AFC.

6.6 A non-exhaustive list of recommended measures for installation of an AFS in niche areas is as follows:

.1 Sea chest: Internal surfaces and inlet gratings of sea chests should be protected by an AFS that is suitable for the flow conditions of the area over the gratings and through the sea chest.

.2 Bow and stern thrusters: Free-flooding spaces which may exist around the thruster tunnel require special attention. The housings/recesses and retractable fittings such as stabilizers and thruster bodies should have an AFC of adequate thickness for optimal effectiveness.

.3 Rudder hinges and stabilizer fin apertures: Rudders and stabilizer fins should be moved through their full range of motion during the coating process to ensure that all surfaces are correctly coated to the specification of the

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SOURCE: MEPC.378(80).pdf
.3 Rudder hinges and stabilizer fin apertures: Rudders and stabilizer fins should be moved through their full range of motion during the coating process to ensure that all surfaces are correctly coated to the specification of the AFC. Rudders, rudder fittings and the hull areas around them should also be adequately coated to withstand the increased wear rates experienced in these areas.

.4 Propeller and shaft: Propellers and immersed propeller shafts are generally not coated but polished. Fouling release coatings or other suitable coatings may be applied where possible and appropriate to maintain efficiency.

.5 Stern tube seal assemblies and the internal surfaces of rope guards: Exposed sections of stern tube seal assemblies and the internal surfaces of rope guards should be carefully painted with AFC appropriate to the degree of water movement over and around these surfaces.

.6 Cathodic protection anodes: Biofouling can be minimized in niche areas if anodes are flush-fitted to the hull, a rubber backing pad is inserted between the anode and the hull or the gap is caulked. Caulking the gap will make the

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SOURCE: MEPC.378(80).pdf
.6 Cathodic protection anodes: Biofouling can be minimized in niche areas if anodes are flush-fitted to the hull, a rubber backing pad is inserted between the anode and the hull or the gap is caulked. Caulking the gap will make the seam or joint watertight. If not flush -fitted, the hull surface under the anode and the anode strap should be coated with an AFC suitable for low water flow to prevent biofouling accumulation. If anodes are attached by bolts recessed into the anode surface, the recess should be caulked to remove a potential niche.

.7 Pitot tubes: Where retractable pitot tubes are fitted, the housing should be internally coated with an AFC suitable for static conditions.

.8 Sea inlet pipes and overboard discharges: Pipe openings and accessible internal areas should be protected by an AFS as far as practicable.
Any anti-corrosive or primer coating used should be appropriate for the specific pipe material and area requirements. Care should be taken in surface preparation and coating application to ensure good adhesion and coating thickness.

6.7 Details for performance monitoring of the AFS should be included in the ship-specific

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SOURCE: MEPC.378(80).pdf
surface preparation and coating application to ensure good adhesion and
coating thickness.

6.7 Details for performance monitoring of the AFS should be included in the ship-specific Biofouling Management Plan (BFMP) and be based on recommendations from the manufacturer of the AFS. Necessary measures to ensure that the AFS remains effective over the specified docking interval, plus any recommendations on how to return the AFS to optimal performance, should be included.

6.8 Manufacturers of AFS are also encouraged to provide information on appropriate cleaning methods, details of maintenance or upgrade protocols specific to the AFS and details on inspection and repair to ensure the effectiveness of their products. Such details are encouraged to be included in the ship-specific BFMP.
RESOLUTION MEPC.378(80) (adopted on 7 July 2023)
2023 GUIDELINES FOR THE CONTROL AND MANAGEMENT OF SHIPS' BIOFOULING TO MINIMIZE THE TRANSFER OF INVASIVE AQUATIC SPECIESMEPC 80/17/Add.1
Annex 17, page 13

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Reinstalling, reapplying or repairing the AFS

6.9 Reinstalling, reapplying or repairing the AFS should be in accordance with

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SOURCE: MEPC.378(80).pdf
Annex 17, page 13

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Reinstalling, reapplying or repairing the AFS

6.9 Reinstalling, reapplying or repairing the AFS should be in accordance with manufacturer's guidance that includes measures for surface preparation to facilitate good adhesion and durability.

6.10 Positions of dry-docking blocks and supports should be varied at each dry -docking, or alternative arrangements made to ensure that areas under blocks are painted with an AFC, at least at alternate dry -dockings. Where it is not possible to alternate th e position of dry-docking support strips, these areas should be specially considered and managed by other means, e.g. the application of specialized coatings or procedures or measures for such areas based on the past arrangement of dry-docking support strips to shift their position step by step for each docking.

6.11 Reinstalling or repairing the MGPS in niche areas should be in accordance with manufacturer's guidance.

6.12 When reinstalling, reapplying or repairing AFS in niche areas, the list of recommended

items in paragraph 6.6 should be considered. A non -exhaustive list of some additional

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SOURCE: MEPC.378(80).pdf
manufacturer's guidance.

6.12 When reinstalling, reapplying or repairing AFS in niche areas, the list of recommended items in paragraph 6.6 should be considered. A non -exhaustive list of some additional recommended measures for reinstallation or reapplication of an AFS in niche ar eas is as follows:

.1 bow and stern thrusters the body and area around bow, stern and any other thrusters prone to coating damage should be routinely maintained during dry-dockings;

.2 recesses within rudder hinges and behind stabilizer fins need to be carefully and effectively cleaned and recoated during maintenance dry-dockings; and

.3 gratings located in sea chests may require a major -refurbishment type of surface preparation at each dry-docking to ensure coating durability.

7 CONTINGENCY ACTION PLANS